

INTRODUCTION TO ANESTHESIA

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What is Anesthesia?



Making
Surgery
Painless



DEFINING ANESTHESIA

- ✓ Anesthesiology is a young specialty historically, especially when compared with surgery or internal medicine.
- ✓ The name anesthesia (Greek *an*, “without,” and *esthesia*, “sensation”).
- ✓ Anesthesia means "loss of sensation".
- ✓ Medicines that cause anesthesia are called anaesthetics.
- ✓ Holmes propose use of the term anesthesia to describe the state that incorporates ;Amnesia, Analgesia, and Hypnosis to make painless surgery possible.
- ✓ Anesthesia is the use of medicine to prevent the feeling of pain or another sensation during surgery or other procedures that might be painful.
- ✓ Anesthesia is a state of temporarily loss of controlled sensation and awareness that is induced for medical purpose.

INTRODUCTION

- ❖ Anesthesia is essential for the practice of surgery
- ❖ Anesthetists receives life of patients in her or his hands
- ❖ It is a duty to administer the safest anesthetic possible, to protect the patients and to give full attention for the anesthetized patient
- ❖ To carry out the responsibility every anesthetist needs a foundation of medical science ,the principle of anesthesia and a sound knowledge of physiology and pharmacology is essential
- ❖ Anesthetists are like no other medical personnel:
- ❖ We are experts at controlling the airway and at emergency resuscitation

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- ❖ we are real-time cardiopulmonologists achieving hemodynamic and respiratory stability for the anesthetized patient;
- ❖ we are pharmacologists and physiologists, calculating appropriate doses and desired responses;
- ❖ we are expert of postoperative care and patient safety
- ❖ we are internists performing pre and post anesthetic medical evaluations
- ❖ we are the pain experts across all medical disciplines and apply specialized techniques in pain clinics and labor wards;

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- we manage the severely sick and injured in critical care units
- we are neurologists, selectively blocking sympathetic, sensory, or motor functions with our regional techniques
- we are trained researchers exploring scientific mystery and clinical phenomenon..
- Anesthesiology is an amalgam of specialized techniques, equipment, drugs, and knowledge that like the growth rings of a tree have built up over time.
- Current anesthesia practice is the summation of the individual efforts and un anticipated discoveries of centuries

THE ROLE ANESTHETIST

The two main activities of the anesthesia provider are

1. Life saving
2. Alleviate suffering

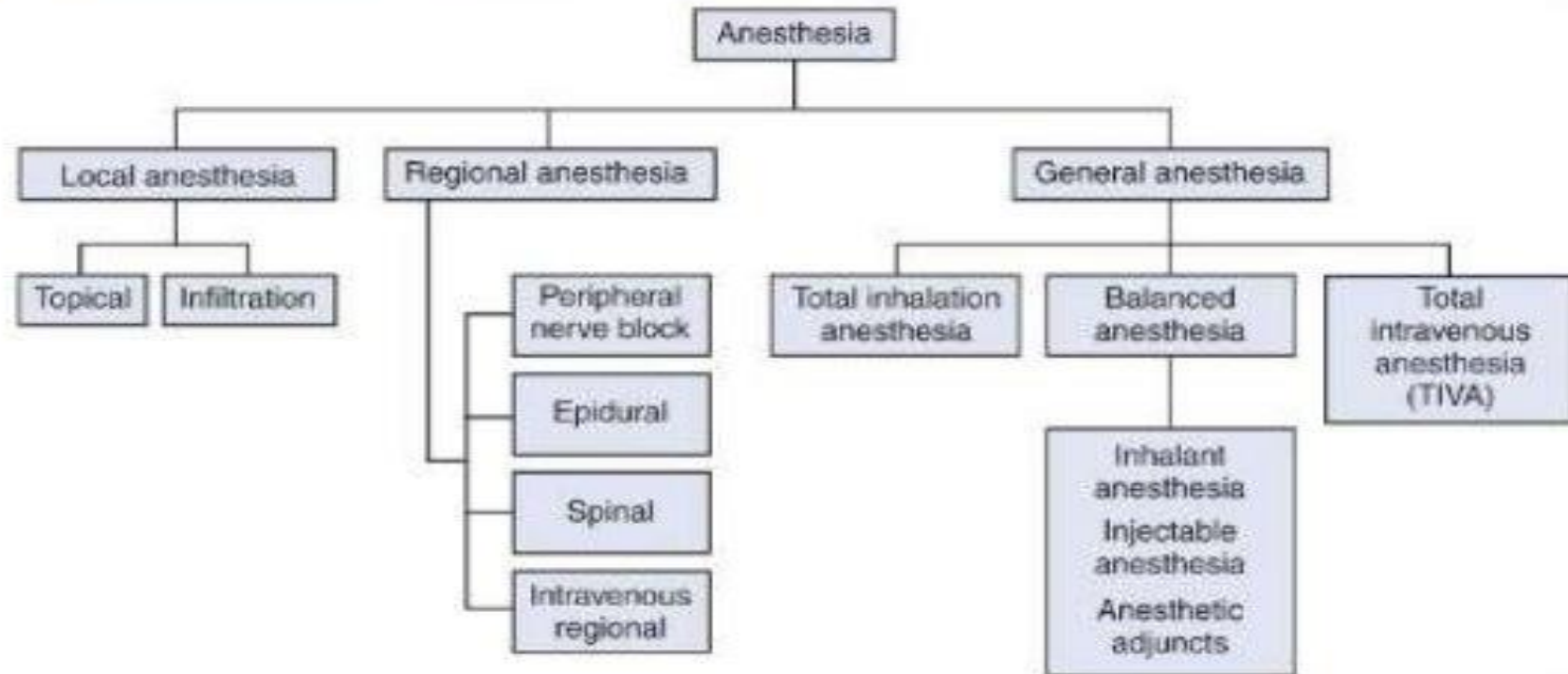
The main activities of the anesthesia provider :-

- ✓ providing anesthesia and pain management to patients before, during and after operations and surgical procedures.
- ✓ Functioning as a guardian of the patient
- ✓ The anesthesia provider should observe the principles of aseptic technique

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- ✓ Monitoring vital functions and parameters
- ✓ Fluid and electrolyte administration
- ✓ Administering anesthetic agent/anesthesia
- ✓ Maintaining anesthesia at the required levels
- ✓ Managing untoward reactions to anesthesia throughout the surgical procedure

Types



TYPES OF ANESTHESIA

There are three types of anesthesia .

I. General anesthesia

II. Regional anesthesia

III. Local anesthesia

I .General anesthesia ;is the induction of a state of unconsciousness with the absence of pain sensation over the entire body, through the administration of anesthetic drugs.

It is used during certain medical and surgical procedures.



I. GENERAL ANESTHESIA

- ❖ General anesthesia acts primarily on the central nervous system to make the patient unconscious and unaware.
- ❖ It is administered via the patient's circulatory system by a combination of inhaled gas and injected drugs.
- ❖ After the initial injection, anesthesia is maintained with inhaled gas anesthetics and additional drugs through an intravenous line (IV).
- ❖ It can broadly be defined as a drug-induced reversible depression of the central nervous system (CNS) resulting in the loss of response to and perception of all external stimuli.

WHERE IN THE CENTRAL NERVOUS SYSTEM DO ANESTHETICS WORK?

- ❖ The biochemical mechanism of action of general anesthetics is not well understood.
- ❖ To induce unconsciousness, anaesthetics have multiple sites of action and affect the central nervous system (CNS) at multiple levels.
- ❖ Common areas of the central nervous system whose functions are interrupted or changed during general anesthesia include the cerebral cortex, thalamus, reticular activating system, and spinal cord.
- ❖ Current theories on the anaesthetized state identify not only target sites in the CNS but also neural networks and loops whose interruption is linked with unconsciousness.
- ❖ Potential pharmacologic targets of general anesthetics are GABA, glutamate receptors, voltage-gated ion channels, and glycine and serotonin receptors.

General anesthesia

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graph TD; A[General anesthesia] --> B[need for unconsciousness<br/>'Amnesia-hypnosis']; A --> C[need for analgesia<br/>'Loss of sensory and autonomic reflexes']; A --> D[need for muscle relaxation];
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need for
unconsciousness
'Amnesia-hypnosis'

need for analgesia
'Loss of sensory and
autonomic reflexes'

need for muscle
relaxation

COMPONENTS OF GENERAL ANESTHETIC STATE



COMPONENT OF GENERAL ANESTHESIA

1. Hypnosis(Unconsciousness) : Unconsciousness is a state which occurs when the ability to maintain an awareness of self and environment is lost.

It involves a complete or near-complete lack of responsiveness to people and other environmental stimuli.

2. Amnesia :Amnesia can be caused temporarily by the use of various sedatives and hypnotic drugs.

Essentially, amnesia is loss of memory.



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3. ***Analgesia***: An analgesic or painkiller is any member of the group of drugs used to achieve analgesia, relief from pain.

Analgesic drugs act in various ways on the peripheral and central nervous systems

4. ***Immobility*** ;loss of motor reflex

5. *Decrease the autonomic responses to noxious stimulation.*

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HYPNOSIS

Examples of Iv drugs ;

- Propofol
- Thiopentone
- Etomidate
- Benzodiazepines
- Ketamine....

Examples of Inhalational drugs; Nitrous oxide(weak) ,Isoflurane, Sevoflurane, Desflurane, Halothane...



ANALGESIA

- Good analgesia (good anesthesia)

Opiates

- Gold standard ,morphine
 - Derivatives diamorphine and codeine
 - Synthetic agents
 - Pethidine
 - Fentanyl/Alfentanil ;short acting
 - Remifentanil ;ultra short acting
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NSAIDS

Gold standard Meloxicam same family of aspirin, ibuprofen and naproxen.

- Diclofenac
- Paracetamol
- ibuprofen
- Cox2 inhibitor

Local anesthetics ;Lidocaine ,Bupivacaine ,Tetracaine

MUSCLE RELAXATION

- Aids intubation
- Helps the procedure(surgery)
- Surgery for long duration

Two types

- Depolarizing and non depolarizing
- Depolarizing muscle relaxant ;E.g. succinylcholine
- Non depolarizing muscle relaxant ; E.g. Vecuronium, pancronium , atracurium,...

STAGES OF GENERAL ANESTHESIA

Stage I: Amnesia and analgesia

This period begins with induction of anesthesia and continues to loss of consciousness.

Stage II: Delirium

This period characterized by not reserved excitation and potentially injurious responses to noxious stimuli, including vomiting, tachycardia, hypertension and uncontrolled movement. The pupils are often dilated

Stage III: Surgical anesthesia

In this target depth for anesthesia, the gaze is central, pupils are constricted, and respirations are regular.

Stage IV: Over dosage

Commonly described as too deep.



FOUR MAIN STAGES ARE RECOGNIZED BASED UPON ;

- Patients body movement
- Respiratory rhythm
- Oculomotor reflex
- Muscle tone

STAGE ONE(AMNESIA AND ANALGESIA STAGE)

- ✓ Conscious and rational ,how ever perception of pain is diminished
- ✓ From beginning of the anesthetic to the loss of consciousness
- ✓ During this stage the patient will demonstrate the following

Able to open eyes on command

Breath normally

Maintain protect reflexes

Tolerate mild painful stimuli

STAGE TWO(DELIRIUM STAGE)

- ❖ Start with the loss of consciousness, ends with the appearance of regular pattern and loss of eyelid reflex
- ❖ Excitement is noted
- ❖ Children will often exhibit this stage during an inhaled induction

The patient may demonstrate the following during this stage;

Movement of limbs and tense struggle

Irregular breathing,

Breath holding

Pupils become dilated but reactive to light

STAGE THREE (SURGICAL ANESTHESIA)

- ❖ This stage begins resumption of regular breathing pattern and ends at the cessation of respiration
- ❖ Surgery generally occurs during one of the four levels
- ❖ No response to surgical incision

LEVEL ONE

The patient may exhibit the following signs during this level

- ❖ Pupillary dilatation may occur but the pupils will become smaller
- ❖ Absence of pupillary reaction to light
- ❖ Eyelid reflex disappears
- ❖ Respiration is regular
- ❖ Vomiting reflex abolished



LEVEL TWO

The patient may exhibit the following signs during this level;

- ✓ Eyes became fixed in the midline
- ✓ Decrease in the activity of the IC and thoracic muscles during respiration
- ✓ Laryngospasm reflex disappears
- ✓ General muscle tone becomes more flaccid

LEVEL THREE

The patient exhibit the following signs during this level;

- ✓ ICM activity decrease to the point of cessation
- ✓ Respiration may come from only the diaphragm
- ✓ Pupils become more dilated

LEVEL FOUR

The patient may exhibit the following signs during this level

- Paralysis of ICM
- Cessation of spontaneous respiration
- pupils become very dilated

STAGE FOUR (ANESTHETIC OVER DOSE)

This stage represent an overdose

- Action must be taken to decrease the inhaled anesthetic or cardiac arrest may occur
- The patient may exhibit the following signs

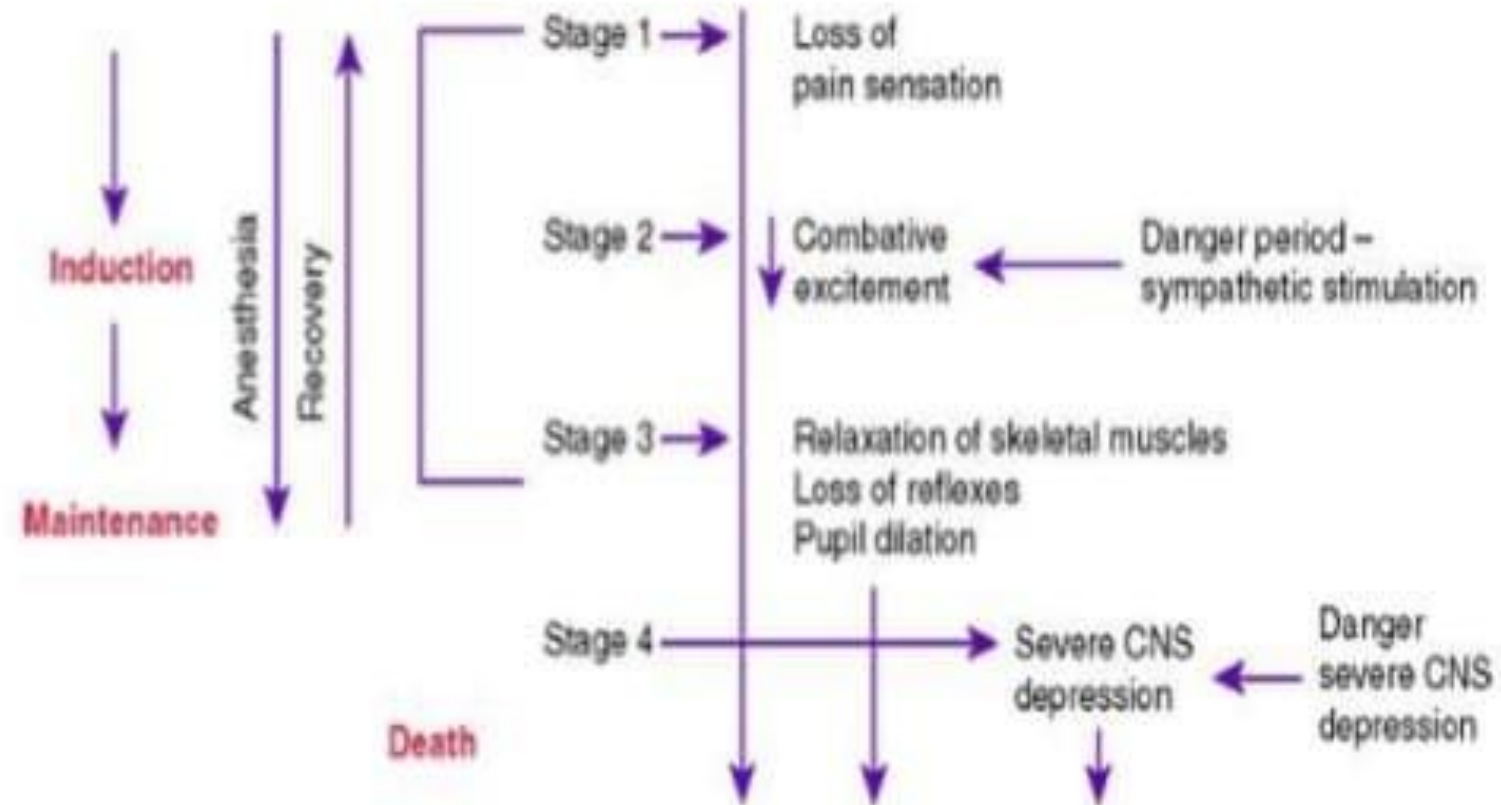
Cessation of spontaneous respiration

Severe bradycardia and hypotension

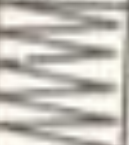
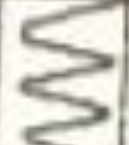
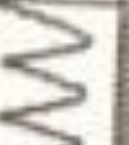
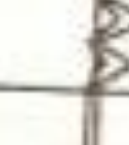
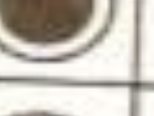
Cardiac arrest

Absence of all reflexes

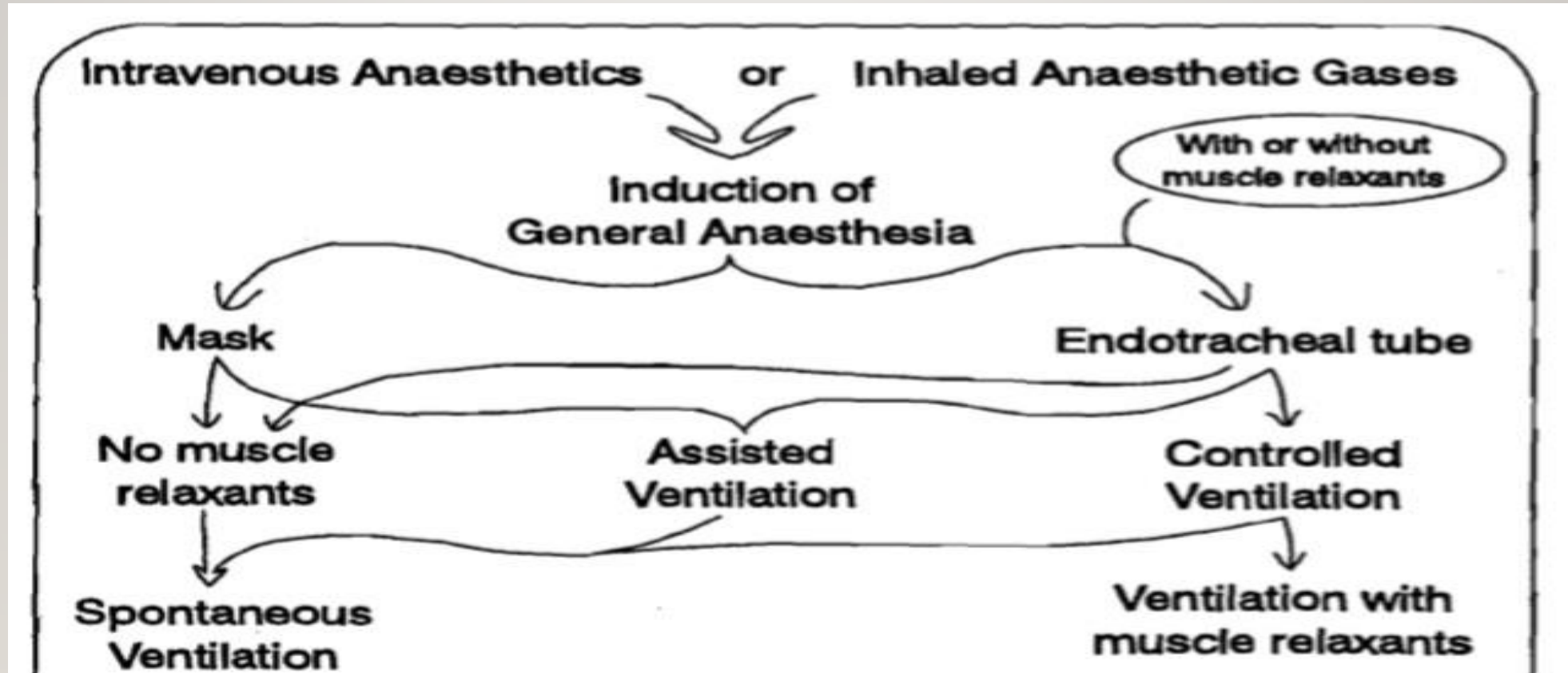




Section

STAGE	RESPIRATION		OCULAR MOVEM.	PUPIL SIZE	REFLEXES	SK. MUS. TONE	B. P.	H.R.	USES
	Thor.	Abd.							
I ANALGESIA			NORMAL						Labour, Incisions & Minor ops.
II DELIRIUM									NIL
III	1								Most of the surgical operations
	2								
	3								
	4								
IV MEDULLARY PARALYSIS									Occasionally reached now NEVER ATTEMPTED

ESTABLISHMENT OF GENERAL ANESTHESIA



ANESTHETIC PROCEDURE SHOULD ;

- ❖ Abolish pain

- ❖ Be completely reversible

- ❖ Be safe !; Comfort _important

Safety _essential and must come first

- ❖ Provide good operating conditions

- ❖ Be acceptable to the patient

ANESTHESIA EVENTS PRACTICALLY

- ❑ Induction; it is the period of time which begins with the administration anesthetic up to the development of surgical anesthesia and done by anesthetic agents
- ❑ Maintenance ;sustaining the state of anesthesia done by inhalational agents...
- ❑ Recovery ;anesthetics stopped at the end of the procedure and consciousness regains

II. REGIONAL ANESTHESIA

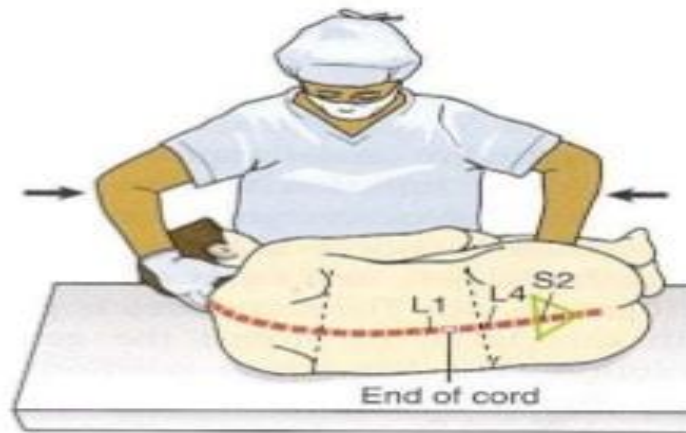
- ❑ **Regional Anesthesia** - involves injection of a local anesthetic around major nerves or the spinal cord to block pain .
- ❑ Is medicine given to temporarily stop the sense of pain in a particular area of the body.
- ❑ A patient remains conscious during a regional anesthetic.

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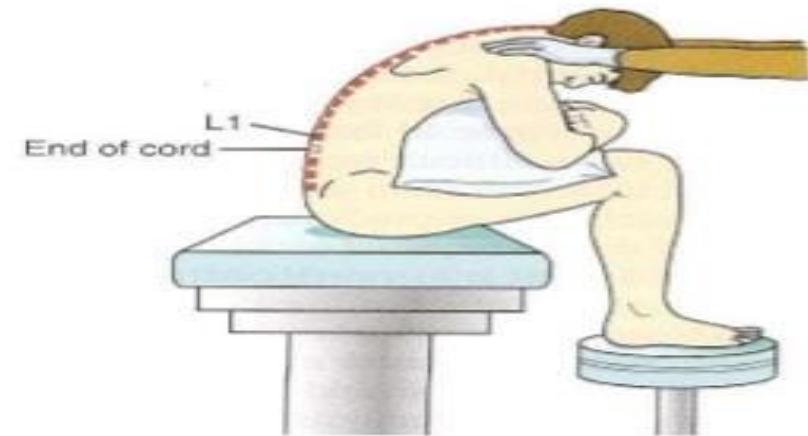
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- ❖ Major types of regional anesthesia include: Central neural blockade refers to either **epidural or spinal anesthesia**.
 - ❖ **Spinal anesthesia**- often used for lower abdominal, pelvic, rectal, or lower extremity surgery.
 - ❖ This type of anesthetic involves injecting a single dose of the anesthetic agent directly into the spinal cord in the lower back, causing numbness in the lower body.

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Spinal Block - Position



Spinal block—lateral position.



Spinal block—sitting position.

CONTRAINDICATIONS TO SPINAL ANESTHESIA

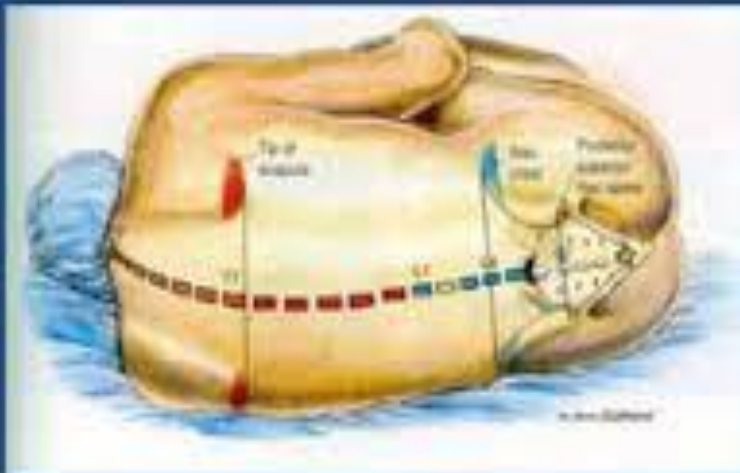
- Patient refusal
- Lack of resuscitative equipment
- Lack of knowledge of procedure
- Coagulopathy
- Previous back surgery (relative)
- Raised intracranial pressure

CONT'D

- ✓ **Epidural** - this anesthetic is similar to a spinal anesthetic and also is commonly used for surgery of the lower limbs and during labor and childbirth.
- ✓ This type of anesthesia involves continually infusing drugs through a thin catheter that has been placed into the space that surrounds the spinal cord in the lower back, causing numbness in the lower body

EPIDURAL ANESTHESIA

EPIDURAL ANESTHESIA



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CAUDAL ANESTHESIA

✓ **Caudal** anesthesia is a common regional technique in pediatric patients.

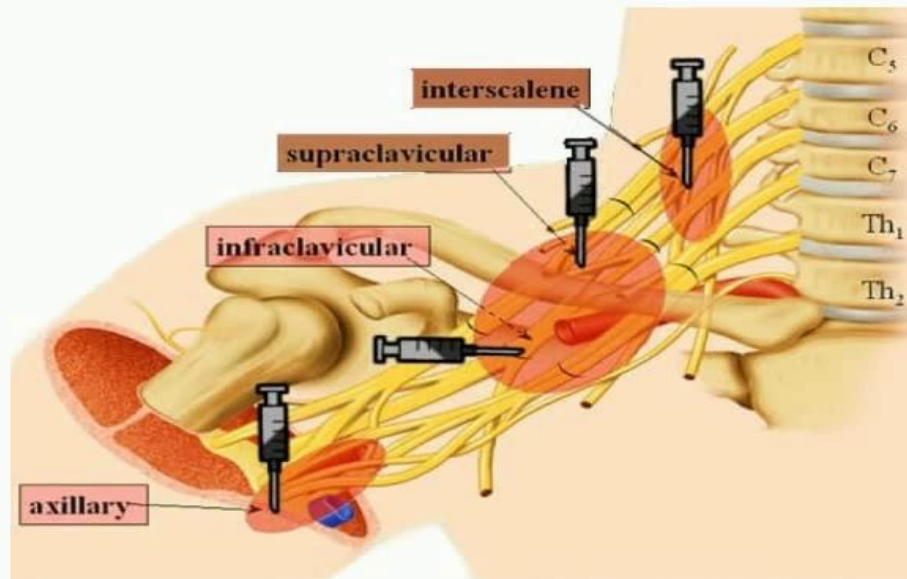


FIGURE 45-20 Positioning an anesthetized child for caudal block and palpation for the sacral hiatus. An assistant gently helps flex the spine.

CONT'D

- ❑ **Peripheral Nerve blocks** - A local anesthetic is injected near a specific nerve or group of nerves to block pain from the area of the body supplied by the nerve
- ❑ A peripheral nerve block is an anesthetic practice used in many surgical procedures. It is accomplished by injecting a local anesthetic near the nerve controlling sensation or movement to the area of the body requiring surgery.
- ❑ Nerve blocks are most commonly used for procedures on the hands, arms, feet, legs, or face.
- ❑ Example - A Brachial Plexus block may be used to provide anesthesia to the entire arm and shoulder.

brachial plexus block



The techniques are classified by the level at which the needle or catheter is inserted for injecting the local anesthetic.

- interscalene block: On the neck.
- supraclavicular block: Immediately above the clavicle.
- infraclavicular block: Below the clavicle.
- axillary block: In the axilla

HOW IT **WORKS**

- In **regional anesthesia**, a numbing medication is injected around the nerves that transmit pain signals from the area involved in the surgery.
- The procedure “blocks” the nerves, ensuring that you will not feel pain during or immediately after surgery.

LOCAL ANESTHESIA

A temporary loss of feeling in one small area of the body caused by special drugs called local anesthetics

- For minor surgery, a local anesthetic can be administered via injection to the site.
- Types are topical, infiltration or injectable
- Topical ;Local application. You can apply local anesthetic ointment to open sores or mouth ulcers. ...
- Infiltration ;Local injection. Your doctor injects a local anesthetic drug under the skin or deeper. ... E.g. for dental extraction ...

GENERAL ANESTHESIA VS. REGIONAL ANESTHESIA

General anesthesia

- ❖ site of action CNS
- ❖ Far greater requirement for intravenous and potentially long-acting medications
- ❖ Intravenous drugs with potential side effects required to achieve same effect
- ❖ Airway management is always necessary and mechanical ventilation is often necessary
- ❖ Patients at risk for hemorrhage

Regional anesthesia

- ❖ Site of action peripheral nerve
- ❖ Very targeted site of drug action
- ❖ Necessitates far less intravenous sedations and analgesia (especially important in elderly patients)
- ❖ Profound analgesia and relaxation
- ❖ No need for airway management or mechanical ventilation
- ❖ Possible decreased blood loss

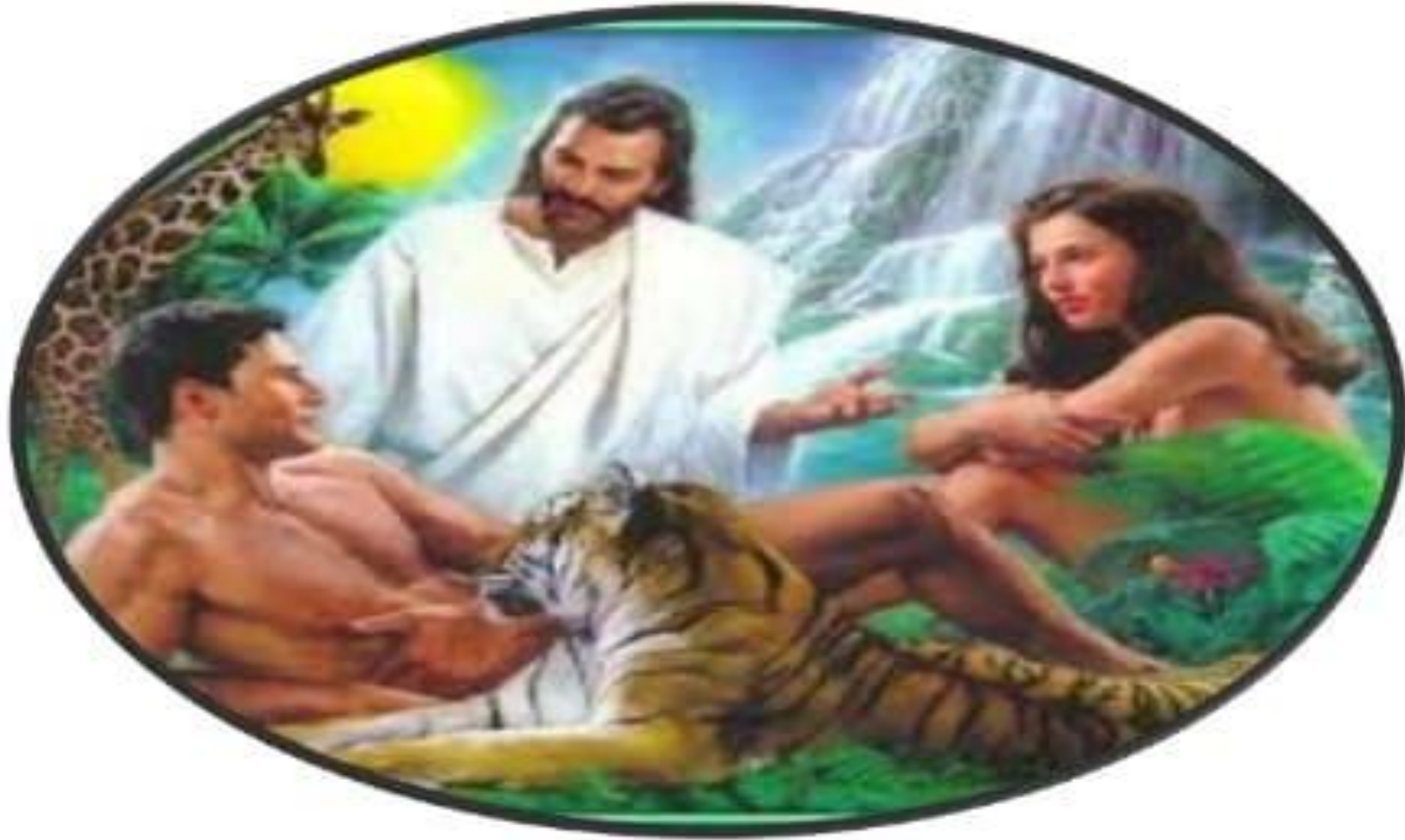
CHOICE OF ANESTHESIA

- ✓ The patients understanding and wishes regarding the type of anesthesia that could be used
- ✓ The type and duration of the surgical procedure
- ✓ The patients mental, physiologic status and stability
- ✓ The presence and severity of coexisting disease
- ✓ The post operative recovery from various anesthesia
- ✓ Option for management of post operative pain
- ✓ Any particular requirement of the surgeon
- ✓ **There is major and minor surgery but only major anesthesia**

COMMON TERMINOLOGIES IN ANESTHESIA

- **Anesthesia** -the absence or abolition of sensation especially that of pain induced by drugs especially as a means of facilitation to surgical procedures .
- **Anesthesiology** –the art and science of rendering a patient insensible to pain by the administration of anesthetic agents and related drugs and procedures
- **An anesthetist** -a qualified HCP who administers anesthetics to produce total or partial loss of sensation in patients during surgical or diagnostic procedures
- **Nurse Anesthetist** -a nurse who specializes in administration of anesthesia
- **Anesthesiologist** -a physician specialist in anesthesia practice. who administers anesthetics to produce total or partial loss of sensation in patients during surgical or diagnostic procedures.

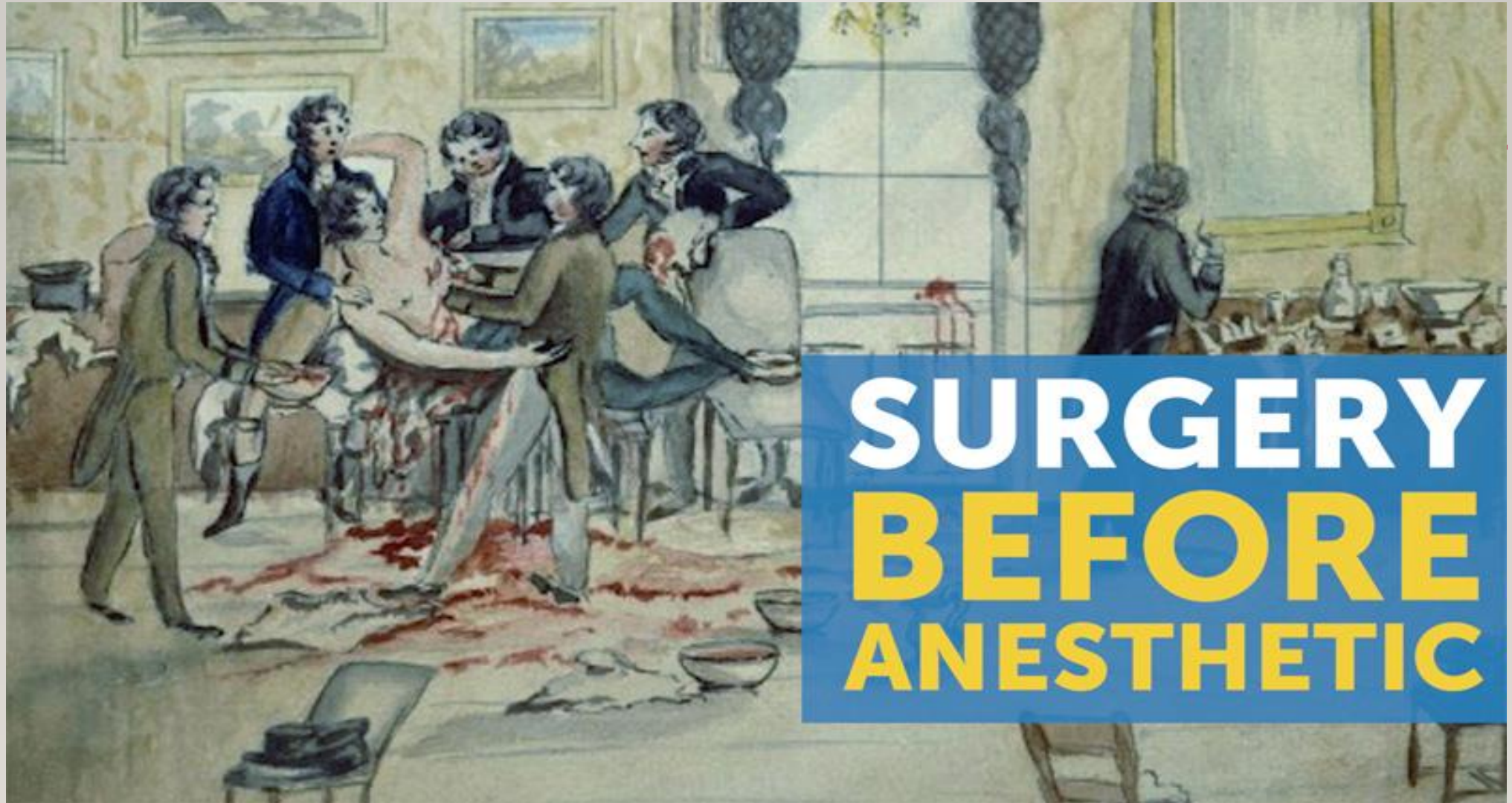
Back To Genesis



PRE 1846 THE FOUNDATION OF ANESTHESIA

.....so the Lord God caused him to fall into a deep sleep. While the man was sleeping, the Lord God took out one of his ribs. He closed up the opening that was in his side.....

Genesis 2:21



**SURGERY
BEFORE
ANESTHETIC**



PRIMITIVE FORMS OF ANESTHESIA

- Phlebotomy
- unconscious
- Regional anesthesia compress nerves
- ischemia
- cold (cryoanalgesia)
- Local anesthesia
- chew coca leaves and spit into the wound

HISTORY OF ANESTHESIA

1846: word 'anesthesia' coined [Holmes] From Greek- an = without -esthesia= sensation

- ✓ Anesthetist are like no other physicians: We are experts at controlling the airway and at emergency resuscitation;
- ✓ we are real-time cardiopulmonologists achieving hemodynamic and respiratory stability for the anesthetized patient;
- ✓ we are pharmacologists and physiologists, calculating appropriate doses and desired responses;
- ✓ we are gurus(expert) of postoperative care and patient safety;
- ✓ we are internists performing perianesthetic medical evaluations

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- ✓ we are the pain experts across all medical disciplines and apply specialized techniques in pain clinics and labor wards;
 - ✓ we manage the severely sick and injured in critical care units;
 - ✓ we are neurologists, selectively blocking sympathetic, sensory, or motor functions with our regional techniques;
 - ✓ we are trained researchers exploring scientific mystery and clinical phenomenon.
 - ✓ Historical examination enables understanding of how these individual components of anesthesia evolved.
 - ✓ Knowledge of the history of anesthesia enhances our appreciation of current practice and foretells where our specialty might be headed

Sushruta 800B.C.
wrote first Surgery textbook



INDIA

The history of anesthesia in India dates back to the era of Susruta, the great surgeon of ancient India. During his time, around 600 bc, operations were performed with the use of opium, wine, and Indian hemp (an herb). Surgical science was called “salya-tantra” (salya: broken parts of an arrow and other sharp weapons; tantra: maneuver



CHINA

- 200AD
- HUA TUO (first person in China to use anesthesia during surgery.
- Major abdominal surgery ;wine and herbs (mafeisan)

ANESTHESIA BEFORE ETHER(PHYSICAL AND PSYCHOLOGICAL ANESTHESIA)

- The Edwin Smith Surgical Papyrus, the oldest known written surgical document, describes 48 cases performed by an Egyptian surgeon .
- Although this remarkable surgical treatise contains no direct mention of measures to lessen patient pain or suffering, Egyptian pictographs from the same era show a surgeon compressing a nerve in a patient's antecubital fossa while operating on the patient's hand.
- Another image displays a patient compressing his own brachial plexus while a procedure is performed on his palm.
- In the 16th century, military surgeon Ambroise Paré became adept at nerve compression as a means of creating anesthesia.

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- Marco Aurelio Severino described the technique of “refrigeration anesthesia” in which snow was placed in parallel lines across the incisional plane such that the surgical site became insensate within minutes.
 - For centuries anatomic dissections were performed only in winter because colder temperatures delayed deterioration of the cadaver.
 - The technique never became widely used, likely because of the challenge of maintaining stores of snow year-round.
 - Formal manipulation of the psyche to relieve surgical pain was undertaken by French physicians Charles Dupotet and Jules Cloquet in the late 1820s with hypnosis, then called **mesmerism**
 - In a well-attended demonstration in 1828, Cloquet removed the breast of a 64-year-old patient while she reportedly remained in a calm, mesmeric sleep.

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- This demonstration made a lasting impression on British physician John Elliotson, who became a leading figure of the mesmeric movement in England in the 1830s and 1840s.
 - Innovative and quick to adopt new advances, Elliotson performed mesmeric demonstrations and in 1843 published numerous Cases of Surgical Operations Without Pain in the Mesmeric State.
 - Support for mesmerism faded when in 1846 renowned surgeon Robert Liston performed the first operation using ether anesthesia in England and remarked, **“This Yankee dodge beats mesmerism all hollow.”**

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- Dioscorides, a Greek physician from the first century AD, commented on the analgesia of mandragora, a drug prepared from the bark and leaves of the mandrake plant.
 - He observed that the plant substance could be boiled in wine, strained, and used “in the case of persons . . . about to be cut or cauterized, when they wish to produce anesthesia.”
 - Mandragora was still being used to benefit patients as late as the 17th century.
 - From the ninth to the 13th centuries, the soporific sponge was a dominant mode of providing pain relief during surgery.

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- Mandrake leaves, along with black nightshade, poppies, and other herbs, were boiled together and cooked onto a sponge.
 - The sponge was then reconstituted in hot water and placed under the patient's nose before surgery.
 - Prior to the hypodermic syringe and routine venous access, ingestion and inhalation were the only known routes for administering medicines to gain systemic effects.
 - Prepared as indicated by published reports of the time, the sponge generally contained morphine and scopolamine—drugs used in modern anesthesia—in varying amounts.

MANDRAKE

*"Give me to drink "mandragora... ..
That I might sleep out this great gap of time"*





OPIUM POPPY

- ❖ The Sumerians were the first to cultivate the poppy.
- ❖ *Papaver somniferous* meaning the poppy of sleep.
- ❖ When the walls of opium poppy are incised a latex like substance is secreted ,from which many of useful products are derived including opium and alkaloid derivatives morphine, codeine ,papaverine..



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- ❖ Alcohol was another element of the pre-ether armamentarium because it was thought to induce stupor and blunt the impact of pain.
 - ❖ Although alcohol is a central nervous system depressant, in the amounts administered it produced little analgesia in the setting of true surgical pain.

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- ❖ In the late 1900s, the manuscript “On the Use of Mafutsuto,” the name given to the anesthetic method by Hanaoka, was translated into English.
 - ❖ Written by Hajime Matsuoka, the manuscript details preanesthetic evaluation, the timing of anesthesia, and the proposed duration of surgery.
 - ❖ The manuscript stated that care should be taken to ensure that lighting is appropriate; therefore it recommended that operations be performed at noon.
 - ❖ Similarly, it stated that operations should not last more than 2 hours, because there was no way to replace intraoperative fluid or blood losses.
 - ❖ The manuscript also contains a section on postoperative care.

INHALED ANESTHETICS

- ✓ For many years nitrous oxide has been known for its ability to induce lightheadedness, and it was often inhaled by those seeking a thrill.
- ✓ It was made by heating ammonium nitrate in the presence of iron filings.
- ✓ The evolved gas was passed through water to eliminate toxic oxides of nitrogen before being stored.
- ✓ **Nitrous oxide** was first prepared in 1773 by Joseph Priestley, an English clergyman and scientist, who ranks among the great pioneers of chemistry.
- ✓ Without formal scientific training, Priestley prepared and examined several gases, including nitrous oxide, ammonia, sulfur dioxide, oxygen, carbon monoxide, and carbon dioxide.

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- ✓ Davy commented that nitrous oxide transiently relieved a severe headache, obliterated a minor headache, and briefly quenched an aggravating toothache.
- ✓ The most frequently quoted passage was a casual entry: “As nitrous oxide in its extensive operation appears capable of destroying physical pain, it may probably be used with advantage during surgical operations in which no great effusion of blood takes place.”
- ✓ This is perhaps the most famous of the “missed opportunities” to discover surgical anesthesia.

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- ✓ Davy's lasting nitrous oxide legacy was coining the phrase “laughing gas” to describe its unique property.
 - ✓ 1844: Colton travelling scientist ,public demonstrations of discoveries gave N₂O to a man bashed shin against stone bench, showed no signs of pain! seen by Wells (dentist from Connecticut) invited Colton to dental practice .
 - ✓ Colton gave N₂O to Wells (wisdom tooth extraction)



ETHER

- ❖ The discovery of surgical anesthetics in the modern era remains linked to inhaled anesthetics.
- ❖ The compound now known as diethyl ether had been known for centuries; it may have been synthesized first by eighth-century .
- ❖ Arabian philosopher Jabir ibn Hayyan or possibly by Raymond Lully, a 13th century European alchemist.
- ❖ But diethyl ether was certainly known in the 16th century, to both Valerius Cordus(perussia) and Paracelsus(swezerland), who prepared it by distilling sulfuric acid (oil of vitriol) with fortified wine to produce an oleum vitrioli dulce (sweet oil of vitriol).

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- ❖ In one of the first “missed” observations on the effects of inhaled agents, Paracelsus observed that ether caused chickens to fall asleep and awaken unharmed.
- ❖ He must have been aware of its analgesic qualities because he reported that it could be recommended for use in painful illnesses.
- ❖ William E. Clarke, a medical student from Rochester, New York, may have given the first ether anesthetic in January 1842. Dental extraction, but not published!
- ❖ From techniques learned as a chemistry student in 1839, Clarke entertained his companions with nitrous oxide and ether.

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- Emboldened by these experiences, he administered ether, from a towel, to a young woman named Hobbie.
 - One of her teeth was then extracted without pain by a dentist named Elijah Pope.
 - However, it was suggested that the woman's unconsciousness was due to hysteria, and Clarke was advised to conduct no further anesthetic experiments.
 - Two months later, on March 30, 1842, Crawford Williamson Long administered ether with a towel for surgical anesthesia in Jefferson, Georgia.

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- ❑ His patient, James M. Venable, was a young man who was already familiar with ether's exhilarating effects, for he reported in a certificate that he had previously inhaled it and was fond of its use.
 - ❑ Venable had two small tumors on his neck but refused to have them excised because he feared the pain that accompanied surgery.
 - ❑ Knowing that Venable was familiar with ether's action, Dr. Long proposed that ether might alleviate pain and gained his patient's consent to proceed.
 - ❑ After inhaling ether from the towel and having the procedure successfully completed, Venable reported that he was unaware of the removal of the tumors.
 - ❑ In determining the first fee for anesthesia and surgery, Long settled on a charge of \$2.00.

WILLIAM MORTON: 'DISCOVERER OF ANESTHESIA'

William T.G. Morton called his creation Letheon, named after the Lethe River of Greek mythology, noted for its waters that helped erase “painful memories.”



WILLIAM MORTON: 'DISCOVERER OF ANESTHESIA'

- ✓ Wells trained William Morton started dental practice together financial failure
- ✓ Morton began experimenting with ether on goldfish , on dog ,on himself
- ✓ 1846: extracted wisdom tooth handkerchief over patient's mouth dripped ether onto it then published the next day.
- ✓ Demonstration: Massachusetts general hospital : patient felt only 'discomfort' Dispute with surgeon
- ✓ Did not receive credit or financial reward for his discoveries
- ✓ Died a poor man (suicide)
- ✓ **GRAVESTONE INSCRIPTION 'Inventor and Revealer of Inhalation Anesthesia: Before Whom, in All Time, Surgery was Agony; By Whom, Pain in Surgery was averted and Annulled; Since Whom, Science has Control of Pain.'**



"Ether Day, 1846" Pictured are Gilbert Abbott (the patient), John Collins Warren, MD (the surgeon), William T. G. Morton (the anesthetist) and Henry J. Bigelow, MD (the junior surgeon).

CHLOROFORM

- James Young Simpson was a successful obstetrician in Edinburgh, Scotland, and among the first to use ether for the relief of labor pain.
- Dissatisfied with ether, Simpson soon sought a more pleasant, rapid-acting anesthetic.
- He and his junior associates conducted a bold search by inhaling samples of several volatile chemicals collected for Simpson by British apothecaries.
- David suggested chloroform, which had first been prepared in 1831.

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- Simpson and his friends inhaled it after dinner at a party in Simpson's home on the evening of November 4, 1847.
- They promptly fell unconscious and, when they awoke, were delighted with their success.
- Simpson quickly set about encouraging the use of chloroform.
- Within 2 weeks, he submitted his first account of its use for publication in The Lancet

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- Chloroform gained considerable notoriety after John Snow used it to deliver the last two children of Queen Victoria.
 - The Queen's consort, Prince Albert, interviewed John Snow before he was called to Buckingham Palace to administer chloroform at the request of the Queen's obstetrician.
 - During the monarch's labor, Snow gave analgesic doses of chloroform on a folded handkerchief.
 - This technique was soon termed chloroform à la reine. (the queen)



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- ✓ Victoria abhorred the pain of childbirth and enjoyed the relief that chloroform provided.
 - ✓ She wrote in her journal, “Dr. Snow gave that blessed chloroform and the effect was soothing, quieting, and delightful beyond measure.”
 - ❖ When the Queen, as head of the Church of England, endorsed obstetric anesthesia, religious debate over the management of labor pain terminated abruptly.
 - ❖ John Snow, already a respected physician, took an interest in anesthetic practice and was soon invited to work with many leading surgeons of the day. In 1848, Snow introduced a chloroform inhaler.



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- ❖ He had recognized the versatility of the new agent and came to prefer it in his practice.
 - ❖ At the same time, he initiated what was to become an extraordinary series of experiments that were remarkable in their scope and for anticipating sophisticated research performed a century later.
 - ❖ Snow realized that successful anesthetics should abolish pain and unwanted movements.
 - ❖ He anesthetized several species of animals with varying strengths of ether and chloroform to determine the concentration required to prevent reflex movement from sharp stimuli.



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- This work approximated the modern concept of minimum alveolar concentration.
 - Snow assessed the anesthetic action of a large number of potential anesthetics but did not find any to rival chloroform or ether.
 - His studies led him to recognize the relationship between solubility, vapor pressure, and anesthetic potency, which was not fully appreciated until after World War II.
 - Snow published two remarkable books, *On the Inhalation of the Vapor of Ether* (1847) and *On Chloroform and Other Anesthetics* (1858).
 - The latter was almost completed when he died of a stroke at the age of 45, and it was published posthumously.

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- In the 19th century, the relief of obstetric pain had significant social ramifications and made anesthesia during childbirth a controversial subject.
- Simpson argued against the prevailing view, which held that relieving labor pain opposed God's will.
- The pain of the parturient was viewed as both a component of punishment and a means of atonement for Original Sin.
- Less than a year after administering the first anesthesia during childbirth, Simpson addressed these concerns in a pamphlet entitled *Answers to the Religious Objections Advanced Against the Employment of Anesthetic Agents in Midwifery and Surgery and Obstetrics*.

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❖ Other volatile agent developments

❖ 1881: cyclopropane

❖ 1951: halothane

❖ 1963: Enflurane

❖ 1965: Isoflurane

BIRTH OF INTRAVENOUS ANESTHESIA

- ❖ Prior to William Harvey's description of a complete and continuous intravascular circuit in *De Motu Cordis* (1628), it was widely held that blood emanated from the heart and was propelled to the periphery where it was consumed.
- ❖ The idea that substances could be injected intravascularly and travel systemically probably originated with Christopher Wren.
- ❖ In 1657, Wren injected aqueous opium into a dog through a goose quill attached to a pig's bladder, rendering the animal "stupefied." wine and cocaine.
- ❖ Wren similarly injected intravenous crocus metallorum, an impure preparation of antimony, and observed the animals to vomit and then die.

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- ❖ Knowledge of a circulatory system and intravascular access spurred investigations in other areas, and Wren's contemporary, Richard Lower, performed the first blood transfusions of lamb's blood into dogs and other animal
 - ❖ In the mid-19th century, equipment necessary for effective intravascular injections was conceived.
 - ❖ Vaccination lancets were used in the 1830s to puncture the skin and force morphine paste subcutaneously for analgesia.
 - ❖ The hollow needle and hypodermic syringe were developed in the following decades but were not initially designed for intravenous use.

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- ❑ In 1845, Dublin surgeon Francis Rynd created the hollow needle for injection of morphine into nerves in the treatment of “neuralgias.”
- ❑ Similarly, Charles Gabriel Pravaz designed the first functional syringe in 1853 for perineural injections.
- ❑ Alexander Wood, however, is generally credited with perfecting the hypodermic glass syringe.
- ❑ In 1855, Wood published an article on the injection of opiates into painful spots by the use of hollow needle and his glass syringe.

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- ❑ In 1872, Pierre Oré of Lyons performed what is perhaps the first successful intravenous surgical anesthesia by injecting chloral hydrate immediately prior to incision.
 - ❑ His 1875 publication describes its use in 36 patients, but several postoperative deaths lent little to recommend this method to other practitioners.
 - ❑ In 1909, Ludwig Burkhardt produced surgical anesthesia by intravenous injections of chloroform and ether in Germany.
 - ❑ Seven years later, Elisabeth Bredenfeld of Switzerland reported the use of intravenous morphine and scopolamine.

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- ✓ The trials failed to show an improvement over inhaled techniques.
- ✓ Intravenous anesthesia found little application or popularity, primarily because of a lack of suitable drugs. In the following decades, this would change.
- ✓ The first barbiturate, barbital, was synthesized in 1903 by Fischer and von Mering.
- ✓ Phenobarbital and all other successors of barbital had very protracted action and found little use in anesthesia.

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- After 1929, oral pentobarbital was used as a sedative before surgery, but when it was given in anesthetic concentrations, long periods of unconsciousness followed.
- The first short acting oxybarbiturate was hexobarbital (Evipal), available clinically in 1932.
- Hexobarbital was enthusiastically received by the anesthesia communities in Europe and North America because its abbreviated induction time was unrivaled by any other technique
- A London anesthetist, Ronald Jarman, found that it had a dramatic advantage over inhalation inductions for minor procedures

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- ✓ Jarman instructed his patients to raise one arm while he injected hexobarbital into a vein of the opposite forearm.
- ✓ When the upraised arm fell, indicating the onset of hypnosis, the surgeon could begin.
- ✓ Patients were also amazed in that many awoke unable to believe they had been anesthetized.
- ✓ Even though the prompt action of hexobarbital had a dramatic effect on the conduct of anesthesia, it was soon replaced by two thiobarbiturates.

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- ✓ In 1932, Donalee Tabern and Ernest H. Volwiler of the Abbott Company synthesized thiopental (Pentothal) and thiamylal (Surital).
- ✓ The sulfated barbiturates proved to be more satisfactory, potent, and rapid acting than were their oxybarbiturate analogues.
- ✓ **Thiopental** was first administered to a patient at the University of Wisconsin in March 1934, but the successful introduction of thiopental into clinical practice followed a thorough investigation conducted by John Lundy and his colleagues at the Mayo Clinic in June 1934.

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- ✓ When first introduced, thiopental was often given in repeated increment as the primary anesthetic for protracted procedures.
 - ✓ Its hazards were soon appreciated. At first, depression of respiration was monitored by the simple method of observing the motion of a wisp of cotton placed over the nose.
 - ✓ Only a few skilled practitioners were prepared to pass a tracheal tube if the patient stopped breathing.
 - ✓ Such practitioners realized that thiopental without supplementation did not suppress airway reflexes, and they therefore encouraged the prophylactic provision of topical anesthesia of the airway beforehand.

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- ❑ The vasodilatory effects of thiobarbiturates were widely appreciated only when thiopental caused cardiovascular collapse in hypovolemic burned civilian and military patients in World War II.
 - ❑ In response, fluid replacement was used more aggressively and thiopental administered with greater caution.
 - ❑ **In 1962, ketamine** was synthesized by Dr. Calvin Stevens at the Parke Davis Laboratories in Ann Arbor, Michigan.
 - ❑ One of the cyclohexylamine compounds that includes phencyclidine, ketamine was the only drug of this group that gained clinical utility.

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- ❑ The other compounds produced undesirable postanesthetic delirium and psychotomimetic reactions.
- ❑ In 1966, the neologism “dissociative anesthesia” was created by Guenter Corrsen and Edward Domino to describe the trancelike state of profound analgesia produced by ketamine.
- ❑ It was released for use in 1970, and although it remains primarily an agent for anesthetic induction, its analgesic properties are increasingly studied and used by pain specialists.

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- **Etomidate** was first described by Paul Janssen and his colleagues in 1964 and originally given the name hypnomidate.
- Its key advantages, minimal hemodynamic depression and lack of histamine release, account for its ongoing utility in clinical practice.
- It was released for use in 1974 and despite its drawbacks (pain on injection, myoclonus, postoperative nausea and vomiting [PONV], and inhibition of adrenal steroidogenesis), etomidate is often the drug of choice for anesthetizing hemodynamically unstable patients.

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- **Propofol, or 2,6-diisopropylphenol**, was first synthesized by Imperial Chemical Industries and tested clinically in 1977.
 - Investigators found that it produced hypnosis quickly with minimal excitation and that patients awoke promptly once the drug was discontinued.
 - In addition to its excellent induction characteristics, the antiemetic action of propofol made it an agent of choice in patient populations prone to nausea and emesis.

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- Regrettably, Cremophor EL, the solvent with which it was formulated, produced several severe anaphylactic reactions, and it was withdrawn from use.
 - Once propofol was reformulated with egg lecithin, glycerol, and soybean oil, the drug reentered clinical practice and gained great success.
 - Its popularity in Britain coincided with the introduction of the LMA, and it was soon noted that propofol suppressed pharyngeal reflexes to a degree that permitted the insertion of an LMA without a need for either muscle relaxants or potent inhaled anesthetics.

LOCAL REGIONAL ANESTHESIA

- ❖ It is the art of rendering a part of the body insensible for surgical operation or manipulation
- ❖ 1855: cocaine isolated from coca plant (Germany)
- ❖ 1860: cocaine purified & named
- ❖ 1884: Koller (Vienna) In 1884, Carl Koller, a resident in ophthalmology in Vienna, was introduced by Sigmund Freud to a new crystalline substance called cocaine. Cocaine, an extract of the coca leaf, was the first effective local anesthetic.
- ❖ After Albert Niemann refined the active alkaloid and named it cocaine, it was used in experiments by a few investigators.

LOCAL ANESTHESIA IN SOUTH AMERICA

- ✓ It was noted that cocaine provided topical anesthesia and even produced local insensibility when injected, but Carl Koller, a Viennese surgical intern, first recognized the utility of cocaine in clinical practice.
- ❖ In 1884, Carl Koller was completing his medical training at a time when many operations on the eye were performed without general anesthesia.
- ❖ Koller sought a local anesthetic to replace ether anesthesia for operations on the eye; because fine suture material to close the eye wound did not exist, any postoperative retching potentially could cause the loss of vision.

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- In 1884, Koller's friend, Sigmund Freud, became interested in the cerebral-stimulating effects of cocaine and gave him a small sample in an envelope, which he placed in his pocket.
 - When the envelope leaked, a few grains of cocaine stuck to Koller's finger and he absentmindedly licked it with his tongue.
 - When his tongue became numb, Koller instantly realized that he had found the object of his search. In his laboratory, he made a suspension of cocaine crystals that he and a laboratory associate tested in the eyes of a frog, a rabbit, and a dog.
 - Satisfied with the anesthetic effects seen in the animal models, Koller dropped the solution onto his own cornea. To his amazement, his eyes were insensitive to the touch of a pin.

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- As an intern, Carl Koller could not afford to attend the Congress of German Ophthalmologists in Heidelberg on September 15, 1884. However, a friend presented his article at the meeting, and a revolution in ophthalmic surgery and other surgical disciplines began.
- Within the next year, more than 100 articles supporting the use of cocaine appeared in European and American medical journals.
- In 1888, Koller immigrated to New York, where he practiced ophthalmology for the remainder of his career.
- American surgeons quickly developed new applications for cocaine. Its efficacy in anesthetizing the nose, mouth, larynx, trachea, rectum, and urethra was described in October 1884.

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- Other regional anesthetic techniques were attempted before the end of the 19th century.
 - The term spinal anesthesia was coined in 1885 by Leonard Corning, a neurologist who had observed Hall and Halsted.
 - Corning wanted to assess the action of cocaine as a specific therapy for neurologic problems.
 - After first assessing its action in a dog, producing a blockade of rapid onset that was confined to the animal's rear legs, he performed a neuraxial block using cocaine on a man .Corning administered one dose without effect, then after a second dose, the patient's legs "felt sleepy."

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- ✓ The man had impaired sensibility in his lower extremity after about 20 minutes and left Corning's office "none the worse for the experience."
- ✓ Although Corning did not describe escape of cerebrospinal fluid (CSF) in either case, it is likely that the dog had a spinal anesthetic and that the man had an epidural anesthetic.
- ✓ No therapeutic benefit was described, but Corning closed his account and his attention to the subject by suggesting that cocainization might in time be "a substitute for etherization in genitourinary
- ✓ Fourteen years passed before spinal anesthesia was performed for surgery.

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- In the interval, Heinrich Quincke of Kiel, Germany, had described his technique of lumbar puncture.
- He offered the valuable observation that it was most safely performed at the level of the third or fourth lumbar interspace because entry at that level was below the termination of the spinal cord.
- Quincke's technique was used in Kiel for the first deliberate cocainization of the spinal cord in 1899 by his surgical colleague, August Bier.
- Six patients received small doses of cocaine intrathecally, but because some cried out during surgery and others vomited and experienced headaches, Bier considered it necessary to conduct further experiments before continuing this technique for surgery.

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- ✓ Professor Bier permitted his assistant, Dr. Hildebrandt, to perform a lumbar puncture, but after the needle penetrated the dura, Hildebrandt could not fit the syringe to the needle, and a large volume of the professor's spinal fluid escaped.
 - ✓ They were at the point of abandoning the study when Hildebrandt volunteered to be the subject of a second attempt.
 - ✓ Their persistence was rewarded with an astonishing success.
 - ✓ Twenty-three minutes after the spinal injection, Bier noted: "A strong blow with an iron hammer against the tibia was not felt as pain. After 25 minutes: Strong pressure and pulling on a testicle were not painful."
 - ✓ They celebrated their success with wine and cigars.

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- That night, both developed violent headaches, which they attributed at first to their celebration.
- Bier's headache was relieved after 9 days of bed rest.
- Hildebrandt, as a house officer, did not have the luxury of continued rest.
- Bier postulated that their headaches were a result of the loss of large volumes of CSF and urged that this be avoided if possible
- The high incidence of complications following lumbar puncture with wide-bore needles and the toxic reactions attributed to cocaine explain his later loss of interest in spinal anesthesia

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- Surgeons in several other countries soon practiced spinal anesthesia, and progress occurred by many small contributions to the technique.
- Theodor Tuffier published the first series of spinal anesthetics from France and later counseled that the solution should not be injected before CSF was seen.
- The first American report was by Rudolph Matas of New Orleans, whose first 94 patient developed postanesthetic meningismus, a frequent complication that was overcome in part by the use of hermetically sealed sterile solutions recommended by E.W. Lee of Philadelphia and sterile gloves as advocated by Halsted.

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- During 1899, Dudley Tait and Guidlo Caglieri of San Francisco performed experimental studies in animals and therapeutic spinals for orthopedic patients.
 - They encouraged the use of fine needles to lessen the escape of CSF and urged that the skin and deeper tissues be infiltrated beforehand with local anesthesia.
 - This had been suggested earlier by William Halsted and the foremost advocate of infiltration anesthesia, Carl Ludwig Schleich of Berlin.

CONT'D

- 1901: Sicard & Cathelin (Paris) first caudal anesthetic
- 1921: Miravé (Spain) first lumbar epidural
- 1908: Also developed Bier block
- Galen : 1020 : endotracheal intubation
- 1543 : intubation and ventilation of animals (Vesalius) Porcine synthesized in 1904
- Additional local anesthetic subsequently introduced clinically include dibucaine (1930) tetracaine (1932) lidocaine (1947) chlorprocaine (1955) mepivacaine (1957)
- Prilocaine (1960) bupivacaine (1963) and etidocaine (1972)

MUSCLE RELAXANT

- ✓ Muscle relaxants entered anesthesia practice nearly a century after inhalational anesthetics .
- ✓ Curare, the first known neuromuscular blocking agent, was originally used in hunting and tribal warfare by native peoples of South America.
- ✓ 1942: curare first used in surgery (Montreal)
- ✓ The curare's are alkaloids prepared from plants native to equatorial rain forests.
- ✓ These jungle tribes also developed nearly identical methods of delivering the toxin by darts, which, after being dipped in curare, maintained their potency indefinitely until they were propelled through blowpipes to strike the flesh of monkeys and other animals of the treetops.

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- Moreover, the American Indians knew of the juice of a herb that would counteract the effects of the poison if administered in time.
 - The earliest clinical use of curare in humans was to ameliorate the tortuous muscle spasms of infectious tetanus.
 - In 1858, New York physician Louis Albert Sayres reported two cases in which he attempted to treat severe tetanus with curare at the Bellevue Hospital.
 - Both his patients died. Similar efforts were undertaken to use muscle relaxants to treat epilepsy, rabies, and choreiform disorders.
 - Treatment of Parkinson-like rigidity and the prevention of trauma from seizure therapy also preceded the use of curare in anesthesia.

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- The use of Curare was a milestone to anesthesia .
- Curare facilitated tracheal intubation and provide abdominal relaxation
- Suxamethonium was sanitized by bovet 1949 and released in 1951 became standard for facilitating tracheal intubation.
- Until recently suxamethonium remained unparalleled in its onset but its occasional side effect continued to fuel the search for comparable substitute
- Recently introduced agents that come close to this goal include vecuronium ,atracurium ,pipacurium ,doxibrium, rocurinum ,and cis atracurium.

OPIOIDS

- ❖ Morphine was isolated from opium in 1905
- ❖ The concept of balanced anesthesia was introduced in 1926 by lunday
- ❖ Opioids (fentanyl alfentanil sufentanil and alfentanil) prevent patient awareness and suppressing autonomic response during surgery .

DEVELOPMENT OF THE AIR WAY

1732: Scotland ,first successful mouth- -mouth resuscitation (miner)

- 1743: Leveret (France)
- First laryngoscope
- Bent reflective spatulafirst oral endotracheal intubation of human for anesthesia (blind digital technique)
- Abandoned technique (tube dislodge caused death)
- 1920: Magill (London)
- Developed Magill's forceps developed
- 1943: Macintosh (Oxford)
- Developed Macintosh blade
- 1949: Described gum elastic urinary catheter (would later become bougie)

EARLY DEVELOPMENT OF ANESTHESIA MACHINE

- The transition from ether in haler and chloroform soaked handkerchiefs to more sophisticated anesthesia delivery equipment such as laryngoscope ,ETT ,suction machine anesthesia machine....and monitoring equipment's .
- The inaesthetic machine of today is based on the constant improvement and the addition of new and safety improving technologies.
- The first anesthetics were given not from a machine but from vaporizers or inhalers. Morton gave his ether through a simple inhaler.
- A breathing tube was placed in the patient's mouth and valves separated the inspired and expired gases. The open drop technique involved using a folded handkerchief over the patient's face, leading to the development of masks

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- Millions of patients were anaesthetized with open masks.
 - These masks were generally open mesh frames that were covered with cloth; ether or chloroform was dropped on to the cloth.
 - Further development of these masks involved a lip to prevent spillage of liquid onto the patient e.g a Schimmel Busch mask.
 - What we call masks were called face pieces.



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- Anesthetic machines were, and still are, either continuous flow or intermittent device machines.
 - An intermittent device, also known as a demand device, only has a flow of gas with the patient's respiration.
 - These devices were most commonly used in dentistry and obstetrics. The best-known continuous flow machine was the Boyle's machine
 - Until the development of the electronic machine all pneumatic machines worked on the basic principle of the Boyle's machine.
 - The original Boyle's machine was a device made up of two N₂O cylinders and two O₂ cylinders in a box with a water sight flow meter and an ether vaporizer.

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- It had a pressure gauge on the cylinders and fine adjustment reducing valves.
 - There was a spirit flame to prevent obstruction of the valves caused by freezing of water in the gas supplies.
 - It had a pressure gauge on the cylinders and fine adjustment reducing valves.
 - There was a spirit flame to prevent obstruction of the valves caused by freezing of water in the gas supplies.

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- A Boyle's vaporizing bottle was later added to the flow meters and later again the flow meters were changed, via a few permutations, to fixed pressure rotameters.
- Much later, safety developments to the rotameters saw entry of oxygen as the last gas admitted to the back bar so that a leak in the other rotameters can not dilute the oxygen delivered, and the international oxygen knob was introduced with the oxygen knob set forward of all other knobs.

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- The oxygen flowmeter was later linked to the nitrous oxide flow meter such that no less than 25 percent oxygen could be delivered.
- An anti-hypoxic device for use whenever nitrous oxide is administered .
- The development of the machine saw the addition of new advances and safety devices.
- Vaporizers developed from simple bubble through or flow over devices often using technology of other industries (the Goldman vaporizer little more than the bowl from a carburettor as used on a petrol engine)

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- The potential for loss of oxygen introduced the need for an oxygen warning device.
- Unlike earlier anesthetics given via an open drop, the safety of room air was no longer available.
- The development of the oxygen warning device is instructive on how many developments occurred.
- An ideal failure device should be reliant solely on the gas the device is monitoring.
- One of the earliest and common devices was the Bosun alarm.
- As oxygen pressure falls a whistle sounds but the whistle is from the reduced pressure of oxygen opening a valve to N₂O.

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- ☐ It cut off the nitrous oxide a whistle sounded and oxygen was supplied at a reduced rate from a reserve cylinder.
 - ☐ Pneumatic machines all have refinements of this type of system.

Further safety features

- ☐ The pin index system for cylinders has been an Australian standard since 1955
- ☐ The introduction of the ventilator alarm
- ☐ Patient safety valves in the circuit and machine
- ☐ Sleeved index and similar system for gas delivery to the machine
- ☐ Color coding of cylinders and pipelines



The 21 st century-digital
revolution

SCOPE OF ANESTHESIA

- Anesthetist need to be involved, concerned, active participants and leaders in their institution and medical community in order to enhance their practice function and image.
- Anesthetist must participate in operating room (OR) management in their facilities and should play a central leadership role.
- OR scheduling, staffing, utilization, and patient flow issues are complex and anesthetist should work hard both to thoroughly understand and positively influence them.

DEFINITION OF PRACTICE OF ANESTHESIA

1. Assessing, consulting and preparing patients to anesthesia.
2. The Anesthetist primary responsibility is to ensure patients' comfort and safety when they are exposed to surgery.
3. Monitoring and restoring homeostasis in preoperative and critically ill patients.
4. Evaluating respiratory function and applying respiratory therapy.

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5. Managing and teaching of cardiac and pulmonary resuscitation.
 6. Teaching, supervising and evaluating the performance of medical and paramedical personnel involved in anesthesia, respiratory care and critical care.
 7. Conducting research at the basic and clinical science and also involvement in the administration of hospitals and medical schools.

